

BIRLA INSTITUTE OF TECHNOLOGY, MESRA, RANCHI
(MID SEMESTER EXAMINATION MO/2023)

CLASS: B.TECH / IMSC
BRANCH: ALL / PHYSICS

SEMESTER : I
SESSION : MO/2023

SUBJECT: MA103 MATHEMATICS-I

TIME: 02 Hours

FULL MARKS: 25

INSTRUCTIONS:

1. The question paper contains 5 questions each of 5 marks and total 25 marks.
 2. Attempt all questions.
 3. The missing data, if any, may be assumed suitably.
 4. Tables/Data handbook/Graph paper etc., if applicable, will be supplied to the candidates
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|---|-----|-----|----|
| Q.1(a) Discuss the convergence of the sequence $\langle \frac{n}{n^2+1} \rangle$. | [2] | CO | BL |
| | | CO1 | 1 |
| Q.1(b) Test the convergence of the series:
$\sum_{n=1}^{\infty} \sqrt{\frac{n}{2(n+1)}}.$ | [3] | CO1 | 2 |
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| Q.2(a) Using the Cauchy's integral test, examine the convergence of the series: $\sum_{n=1}^{\infty} n e^{-n^2}$. | [2] | CO1 | 2 |
| Q.2(b) Apply the Leibnitz's test, check the convergence of the series:
$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{3n-2}.$ | [3] | CO1 | 2 |
| | | | |
| Q.3(a) Find the rank of the matrix:
$\begin{pmatrix} 1 & 2 & 3 \\ 1 & 4 & 2 \\ 2 & 6 & 5 \end{pmatrix}.$ | [2] | CO2 | 2 |
| Q.3(b) Test the consistency of the system of equations:
$2x - 3y + 7z = 5; 3x + y - 3z = 13; 2x + 19y - 47z = 32.$ | [3] | CO2 | 3 |
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| Q.4 Find the eigenvalues and eigenvectors of the matrix:
$\begin{pmatrix} 3 & 1 & 4 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{pmatrix}.$ | [5] | CO2 | 3 |
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| Q.5(a) Show that limit of the function:
$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2-y^2}{x^2+y^2} \text{ does not exist.}$ | [2] | CO3 | 2 |
| Q.5(b) Find the partial derivatives $f_x(0,0)$ and $f_y(0,0)$ of the function:
$f(x,y) = \frac{x^3-y^3}{x^2+y^2}, (x,y) \neq (0,0).$ | [3] | CO3 | 2 |

:::::13/10/2023:::::